



# Amazon Bedrock Inference Profiles

A logical enterprise layer between your application and the foundation model – enabling cost tracking, routing, monitoring, and governance without changing your application code.

☁ AMAZON BEDROCK

ENTERPRISE MODEL MANAGEMENT

# What Is an Inference Profile?

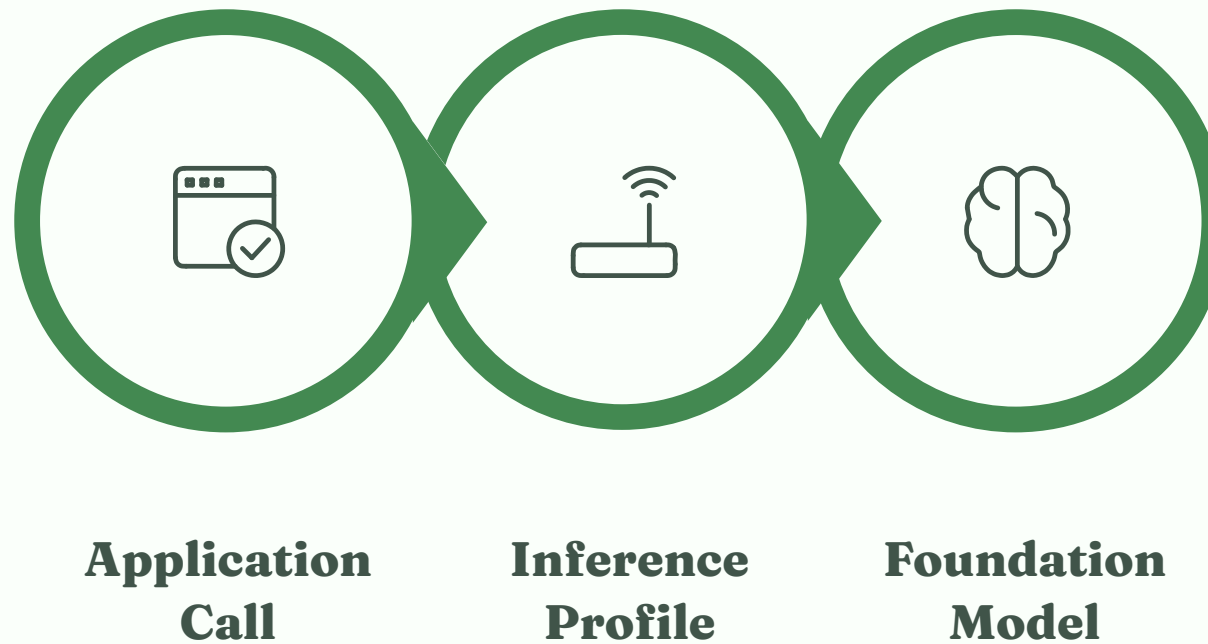
An **Inference Profile** is an Amazon Bedrock resource that acts as a managed invocation endpoint. Rather than calling a foundation model directly, your application targets the inference profile – which handles routing, tracking, and observability on your behalf.

## Foundation Model

The raw AI engine – a large language model such as Meta Llama or Anthropic Claude, accessed through Amazon Bedrock.

## Inference Profile

The enterprise wrapper around the AI engine. It adds routing logic, cost attribution, usage metrics, and operational controls – all transparently to the calling application.



This abstraction decouples application logic from model configuration, making it dramatically easier to manage AI at enterprise scale.

# Without vs. With Inference Profiles

## ✗ Direct Model Invocation

- Application is tightly coupled to a specific model ID
- No built-in cost attribution or team-level tracking
- Limited application-level monitoring and metrics
- Region-specific configuration must be managed manually
- Switching models requires application code changes

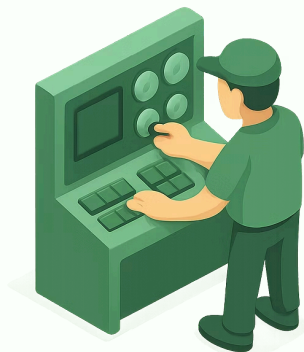
## ✓ Using an Inference Profile

- Application is abstracted from the underlying model
- Granular cost tracking and usage attribution per team or app
- CloudWatch metrics available out of the box
- Cross-Region support via system-defined profiles
- Model changes are transparent to the calling application



# Two Types of Inference Profiles

Amazon Bedrock offers two distinct profile types, each designed for a different operational purpose. Understanding when to use each is key to designing a robust enterprise AI platform.



## Application Inference Profile

**Created by customers.** Designed to track usage, allocate costs, and monitor specific applications or teams. You attach tags (project, department, environment) and point the profile at any supported foundation model. Ideal for chargeback and team-level governance.



## Cross-Region (System-Defined) Profile

**Provided by AWS.** Pre-configured profiles that automatically route requests across multiple AWS Regions to maximize throughput and availability. AWS maintains the list of supported Regions. Ideal for high-traffic workloads and resilience requirements.

# Enterprise Benefits at a Glance

Inference profiles unlock four critical capabilities that organizations need to operate foundation models responsibly and efficiently at scale.



## Cost Allocation

Assign separate inference profiles to each team – Finance, Support, Engineering – to get precise, per-team usage and cost reports directly in AWS Cost Explorer.



## CloudWatch Monitoring

Track request count, latency, token usage, errors, and invocation volume per profile. Surface operational health at the application level without custom instrumentation.



## Tagging & Governance

Attach standard AWS tags – Project, Department, Environment, Business Unit – to each profile for cost governance, compliance reporting, and organizational policy enforcement.



## Centralized Control

Manage model access, routing rules, and cost policies from a single control plane. Enforce IAM-based access control per profile to limit which teams can invoke which models.



# Cross-Region Inference: Scaling Beyond a Single Region

## Without Cross-Region

Your application is pinned to a single Region – for example, **ap-south-1**. During a traffic spike, requests queue for available capacity. There is no automatic failover or load distribution. Throughput is capped by a single Region's quota.

## With Cross-Region Inference

Amazon Bedrock automatically selects an eligible destination Region from the profile – routing across **US East, US West, EU West**, or other supported Regions – to improve throughput and resilience. Two scopes are available:

- **Geographic profiles** – keep routing within a defined geography (US, EU, or APAC) for data residency compliance
- **Global profiles** – maximize available capacity across all supported Regions

# One Line of Code Changes Everything

Adopting inference profiles requires minimal code changes. The only difference is the value passed to `modelId` – the rest of your invocation logic remains identical.

## Direct Model Invocation

```
response = client.converse(  
    modelId="meta.llama3-8b-instruct-v1:0",  
    messages=messages  
)
```

Tightly coupled to a specific model ID. No tracking, no tagging, no routing control.

## Using an Inference Profile

```
response = client.converse(  
    modelId="arn:aws:bedrock:...:application-inference-  
profile/devops-ai",  
    messages=messages  
)
```

The inference profile ARN becomes the invocation target. All enterprise capabilities activate transparently.

- ❏ Only the `modelId` value changes. All other API parameters, authentication, and response handling remain exactly the same – making adoption low-risk and incremental.

# Enterprise AI Platform: Putting It All Together

A mature enterprise AI platform uses a dedicated inference profile for each application or team, all routing through Amazon Bedrock Runtime to a shared pool of foundation models.

## HR Chatbot → HR Profile

Independent usage tracking and cost reporting for the HR organization. Tagged with department=HR.

## DevOps Copilot → DevOps Profile

Separate quota and metrics for engineering tooling. Enables per-team chargebacks and SLA monitoring.

## Customer Support → Support Profile

Isolated observability for customer-facing AI. Latency and error budgets tracked independently from internal tools.

**Key Takeaway:** Foundation models generate intelligence. Inference profiles make those models *manageable at enterprise scale* – adding tracking, governance, routing, and operational visibility without changing application code.